

AI Agents for Kaggle Competitions

An end-to-end LLM agent benchmarked against two frozen yardsticks on 20 Kaggle competitions

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1 · Motivation

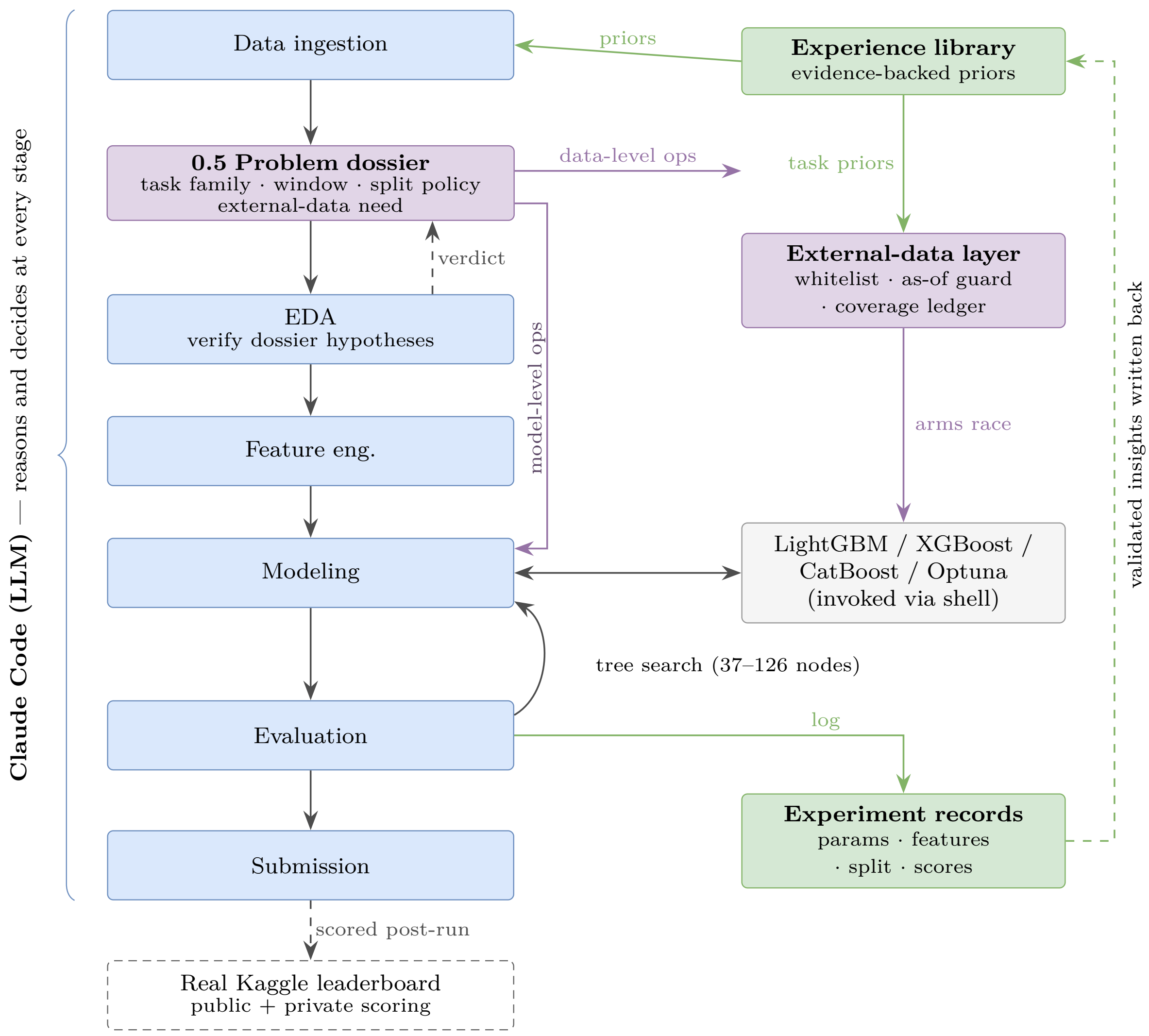
Why these tasks are hard

- **Kaggle competitions** demand the full expert loop — validation design, feature engineering, model selection, ensembling — where the difference between top entries sits at the **4th–5th decimal** of the metric.
- With gaps that small, **local cross-validation is an unreliable compass**: a pipeline’s own validation score routinely disagrees with the real leaderboard, so no local number can be trusted to pick a winner.
- **Data visualization** (EDA, diagnostics, reporting) is essential but repetitive: every dataset needs distribution, drift, and correlation views before modeling can be trusted.

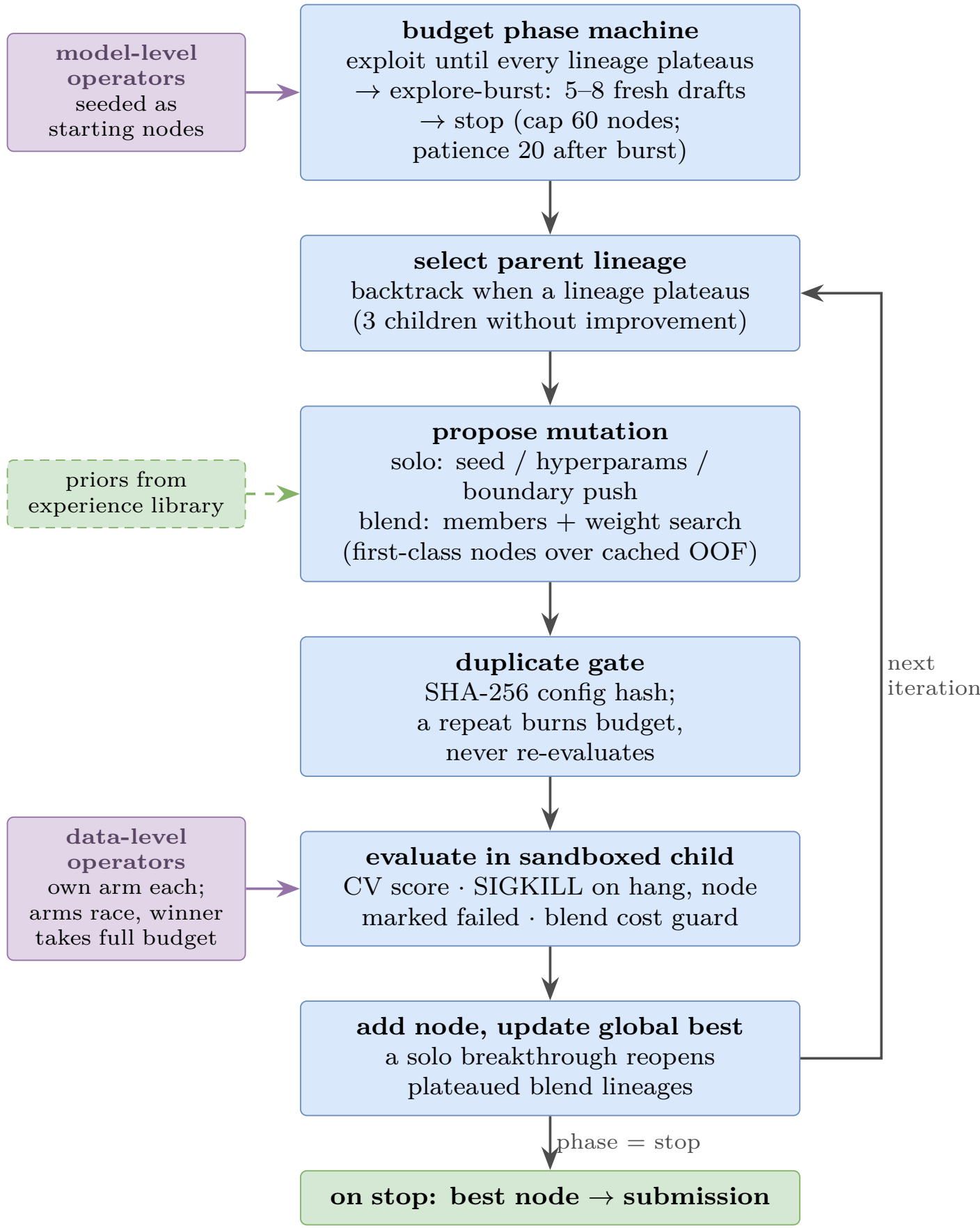
Why an AI agent

- Tireless, systematic search: 37–224 candidate configurations per competition, every experiment logged and reproducible.
- Runs the same disciplined protocol on 20 competitions serially — a human cannot hold this consistency.
- Generates per-dataset EDA figures and per-competition reports automatically, so every claim is grounded in an artifact.

2 · Workflow



The pipeline, read top to bottom: the LLM reasons at every stage; the **Stage-0.5 dossier** (injection layer) classifies the task and injects external data *upstream*, before any modeling; submissions are scored post-run on the real leaderboard. Colour = role: blue LLM pipeline · purple injection layer · green evidence loop · gray Auto-ML tools. Solid arrows = execution flow; dashed = feedback, advisory, or post-run steps.



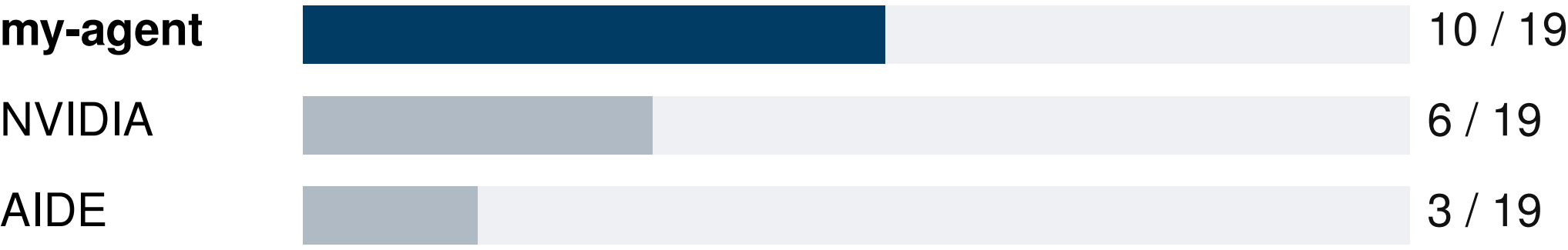
Tree-search loop: ensembles are first-class nodes with OOF caching and weight search; exploit until plateau, then one forced exploration burst; the loop terminates at the best-node submission. Purple = the injection layer’s operator entry points; green = experience-library priors.

3 · Results — Kaggle benchmark

Benchmark standing (scored 2026-08-18, 19 of 20 lanes)

All 20 competitions run under the frozen Stage-0.5 architecture in credential-free sandboxes with a per-lane transcript audit; the 19 completed lanes are scored on the real leaderboard, each duel adjudicated by a paired test on Kaggle’s own public/private split. **my-agent leads the standing — 15 wins / 7 losses (68% excl. ties)** vs. AIDE 44% and NVIDIA 40%, and beats its own development-lane score on 12 of 19. The final lane is re-running clean after an audit discard (TBD).

Best private score — head-to-head count (19 scored)



Private-leaderboard scores, all 20 benchmark competitions

Comp.	Metric	mine	NV	AIDE
s3e1	RMSE ↓	0.55275	0.55550	0.55552
s3e3	AUC ↑	0.87157	0.89537	0.86863
s3e5	QWK ↑	0.57223	0.56974	0.58138
s3e7	AUC ↑	0.90441	0.90458	0.90109
s3e9	RMSE ↓	12.3319	12.2291	12.3067
s3e11	RMSLE ↓	0.29368	0.29624	0.29445
s3e14	MAE ↓	333.186	330.716	332.316
s3e16	MAE ↓	1.33822	1.34001	1.34224
s3e19	SMAPE ↓	49.4088	48.2656	48.3413
s4e11	Acc. ↑	0.94107	0.94076	0.94056

Comp.	Metric	mine	NV	AIDE
s5e1	MAPE ↓	0.08401	0.14594	0.17917
s5e10	RMSE ↓	0.05576	0.05588	0.05579
s6e1	Error ↓	8.71516	8.73239	8.72191
afsis	MCRMSE ↓	0.48917	0.69855	0.49861
cat-in-dat	AUC ↑	0.80243	0.77084	0.80217
conway	MAE ↓	0.10414	0.11189	0.11065
s6e2	AUC ↑	0.95505	0.95508	0.95516
tps-aug	AUC ↑	0.58930	0.58730	0.59106
tps-jan	SMAPE ↓	TBD	4.63651	10.6664
tps-sep	SMAPE ↓	28.4961	5.48351	9.72095

Private-leaderboard score of each frozen submission; **bold** = best of the three; TBD = the lane re-running clean. On s3e19, the canonical failure case, the controlled with/without-layer ablation scores 48.243 against 50.455 without the layer, and all **6 of 6** controlled external-data arms beat their matched baselines.

Mid-complexity three-way (offline-graded, private split)

Competition	Metric	my-agent	NVIDIA	AIDE
us-patent (NLP)	Pearson r ↑	0.86658	0.86160	0.65821
ventilator (time series)	MAE ↓	0.16525	0.40093	0.34591
SIIM-ISIC (imaging)	AUC ↑	0.93529	0.89822	0.88579

4 · Advantages & Difficulties

Expected	Experienced
<i>The agent would quickly beat most humans.</i>	Development runs landed above average, with wins concentrated where no mature public solutions exist; the frozen-architecture re-run now leads the paired standing on the 19 scored lanes.
<i>One run and one score decide a winner.</i>	Score noise and run-to-run variance leave many duels undecided ; a paired error model became mandatory.
<i>The agent follows the rules by default.</i>	A transcript audit caught one lane reading its own competition’s recorded findings — the clean re-run’s local validation scored <i>worse</i> , consistent with the advantage being real — and later caught a lane whose write-back polluted the next lane’s priors.
<i>Fairness is mostly a mindset.</i>	It is infrastructure : isolated data roots, credential-free sandboxes, and per-lane transcript audits cost real engineering time — but without them the numbers are not comparable.

5 · Discussion — advice

- **State the error model with every claim.** A ranking without an error scale is noise: many of our duels are undecidable, and the winner flips when the evaluation set changes.
- **Never trust local CV for between-agent claims.** Score on the real leaderboard; CV describes a pipeline’s internal view only.
- **Freeze your yardsticks.** Improving a baseline mid-study silently turns the benchmark into an optimisation of the baseline.
- **Isolate mechanically, then audit the transcript.** Sandboxes stop leaks you anticipated; reading what the agent actually did catches the ones you didn’t — in both directions (two real contaminations, one auditor false positive).
- **Put knowledge injection upstream.** At the blend stage it was worth $\sim 10^{-5}$ and was abandoned; moved into problem identification it fires on exactly the **2/18** competitions that need external data, lifts the canonical failure case from last to first in the controlled ablation, and all **6 of 6** external-data arms beat their matched baselines.
- **No local signal selects the *form* of injection.** The one surviving selector died on its first pre-registered prediction: the dossier can decide *whether* to inject, but a human still decides *how*.